Skill -3 OS

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <termios.h>
#include <unistd.h>

#define INITIAL_SIZE 16

typedef struct Command
{
    char *text;
    struct Command *next;
} Command;

struct termios original_terminal;

/* Disable terminal's normal line input */
void enable_raw_mode()
{
    struct termios raw;

    tcgetattr(STDIN_FILENO, &original_terminal);
    raw = original_terminal;

    raw.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
```

```c
}

/* Restore normal terminal */
void disable_raw_mode()
{
    tcsetattr(STDIN_FILENO, TCSAFLUSH, &original_terminal);
}

/* Add command to linked-list history */
void add_history(Command **head, const char *text)
{
    Command *new_node = malloc(sizeof(Command));

    if (new_node == NULL)
    {
        perror("malloc");
        exit(1);
    }

    new_node->text = malloc(strlen(text) + 1);

    if (new_node->text == NULL)
    {
        perror("malloc");
        free(new_node);
        exit(1);
```

```c
    }

    strcpy(new_node->text, text);
    new_node->next = NULL;

    if (*head == NULL)
    {
        *head = new_node;
    }
    else
    {
        Command *temp = *head;

        while (temp->next != NULL)
            temp = temp->next;

        temp->next = new_node;
    }
}

/* Count history entries */
int history_count(Command *head)
{
    int count = 0;

    while (head != NULL)
```

Skill -3 OS

```
anirudh@LAPTOP-H9V5A2PN:~/ossp_practical$ gcc skill3.c -o skill3
./skill3
> hello
You entered: hello
> operating systems
You entered: operating systems
>
anirudh@LAPTOP-H9V5A2PN:~/ossp_practical$ nano skill3.c
```